Advanced RAG with Vector Databases and Retrievers

Today's goal • 3/3 items

Today's Skill Points 14 XP

See skill progress

Module 1

Advanced Retrievers for RAG

Welcome to the Course

- ✓ Course Introduction
  Video • 4 min
- ✓ Course Overview
  Reading • 5 min
- ✓ RAG and Agentic AI Professional Certificate Overview
  Video • 5 min
- ✓ Reading: Helpful Tips for Course Completion
  Ungraded Plugin • 5 min

Work with Advanced Retrievers in LangChain

- ✓ Explore Advanced Retrievers in Langchain: Part 1
  Video • 2 min
- ✓ Explore Advanced Retrievers in Langchain - Part 2
  Video • 5 min
- ✓ Lab: Build a Smarter Search with LangChain Context Retrieval
  Ungraded App Item • 1h
- ✓ [Optional] Interactive LangChain Lesson Recap Podcast (AI-Powered)
  Ungraded App Item • 10 min
- ✓ Practice Quiz: Work with Advanced Retrievers in LangChain
  Practice Assignment • Grade: 85.71%

Work with Advanced Retrievers in LlamaIndex

- ✓ Advanced Retrievers in LlamaIndex
  Video • 8 min
- ✓ Lab: Explore Advanced Retrievers in LlamaIndex
  Ungraded App Item • 1h
- ✓ [Optional] Interactive LlamaIndex Lesson Recap Podcast (AI-Powered)
  Ungraded App Item • 10 min
- ✓ Practice Quiz: Work with Advanced Retrievers in LlamaIndex
  Practice Assignment • Grade: 100%

Module Summary and Evaluation

- ✓ Summary and Highlights: Advanced Retrievers for RAG
  Reading • 3 min
- ✓ Reading: Cheat Sheet: Advanced Retrievers for RAG
  Ungraded Plugin • 15 min
- ✓ Graded Quiz: Advanced Retrievers for RAG
  Graded Assignment • Grade: 100%
- ○ [Optional] Meet and Greet
  Discussion Prompt • 2 min

Module 2

Build a Comprehensive RAG Application

Introduction to FAISS for RAG

- ✓ Introduction to FAISS and How it compares to ChromaDB
  Video • 5 min
- ✓ Lab: Semantic Similarity with FAISS
  Ungraded App Item • 1h
- ✓ Reading: Hierarchical Navigable Small World (HNSW)
  Ungraded Plugin • 30 min
- ✓ Lab: AI-Powered YouTube Summarizer, QA Tool with RAG, LangChain, FAISS
  Ungraded App Item • 1h
- ✓ [Optional] Interactive FAISS/HNSW Lesson Recap Podcast (AI-Powered)
  Ungraded App Item • 10 min
- ✓ Practice Quiz: Introduction to FAISS for RAG
  Practice Assignment • Grade: 100%

Module Summary and Evaluation

- ✓ **Summary and Highlights: Build a Comprehensive RAG Application**
  Reading • 2 min
- ○ Cheat Sheet: Build a Comprehensive RAG Application
  Ungraded Plugin • 15 min
- ○ Graded Quiz: Build a Comprehensive RAG Application
  Graded Assignment • 21 min

Course Wrap-Up

- ○ Course Wrap-Up
  Video • 6 min
- ○ Congratulations and Next Steps
  Reading • 2 min
- ○ Team and Acknowledgements
  Reading • 2 min

Summary and Highlights: Build a Comprehensive RAG Application

0:33 / 0:33 1x

Congratulations! You have completed this lesson. At this point in the course, you know:

- FAISS and Chroma DB are built for different goals
- FAISS gives you control over indexing, but lacks metadata and distribution by default
- Chroma DB is easier to deploy and supports metadata, but has fewer indexing options
- You can extend FAISS with Milvus for scalability and metadata
- You can extend both tools with LangChain or LlamaIndex for RAG pipelines
- You should choose the right tool based on your project's size, complexity, and infrastructure

Go to next item ✓ Completed

Ungraded Plugin **Cheat Sheet: Build a Comprehensive RAG Application starts in 2s** Start Pause

Like Dislike Report an issue

Go to next item